
CIPHER: A Benchmark for Inverse Transformation Program Recovery in Multimodal Agents

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Abstract

We introduce **CIPHER**, a procedurally generated benchmark for *inverse transformation program recovery* in tool-using multimodal systems. In CIPHER, a model receives a corrupted video observation together with natural-language instructions describing the transformations required to restore it. The system must (i) infer the latent transformation program, (ii) realize the inverse tool chain via `ffmpeg`, and (iii) localize and extract a short target string that is readable only after successful recovery. This yields a stage-decomposed evaluation of **program inference** (S1), **tool-chain execution** (S2), and **terminal-state extraction** (S3), rather than a single monolithic score.

CIPHER spans five calibrated difficulty levels that vary key visibility, transformation-chain length, instruction ambiguity, and distractor density, with deterministic seed-based replay and procedural regeneration for contamination resistance and reproducible regression testing. We also instantiate a DAVIS-2017 video-temporal split-key domain that measures multi-frame aggregation and operation inversion with dense process partial credit. We evaluate OCR baselines, heuristic pipelines, and frontier multimodal models. Current systems are strong in S3-only extraction when given the correct terminal frame, but remain fragile in end-to-end recovery where temporal localization, transformation sequencing, and executable tool use must all be correct simultaneously. CIPHER is therefore best viewed as a diagnostic benchmark for inverse procedural reasoning in multimodal agents, not merely as another passive video-understanding leaderboard.

1 Introduction

Recent multimodal models are increasingly strong at passive video understanding: they can answer questions about actions, summarize clips, and describe temporal events with surprising fluency. But many agentic failures do not occur at the level of recognizing what is in a video; they occur at the transition from *understanding* to *executable recovery*. A system may understand that a video has been mirrored, reversed, or sped up, yet still fail to realize the correct inverse tool chain, search the right temporal window, or extract the target signal from the restored result.

CIPHER targets exactly this gap. It instantiates a narrow but revealing setting: given a corrupted video observation and natural-language instructions, a system must infer, invert, and execute a latent transformation program, then localize and extract the target signal. The surface form resembles a video-editing challenge, but the underlying capability is *inverse multimodal procedural reasoning* under tool use.

Despite the proliferation of video-capable VLMs [OpenAI, 2023, Google, 2023, Wang et al., 2024], no benchmark systematically tests this compound capability. While current video understanding benchmarks such as Video-MME [Fu et al., 2024] and LongVideoBench [Wu et al., 2024] assess passive temporal reasoning and question answering, they do not require active, sequential manipulation of the visual stream based on language. Conversely, video generation and editing benchmarks

like VBench [Huang et al., 2023] evaluate perceptual quality rather than the precise execution of rigid transformational rules. Neither paradigm captures the tight coupling of instruction-following, procedural tool use, and visual extraction required here.

Two design choices make CIPHER deliberately interpretable. First, the transformation vocabulary is intentionally minimal and reversible: `reverse`, `hflip`, `vflip`, `rotate90`, `rotate270`, and `speed2x`. This keeps failures attributable to reasoning and execution rather than to an uncontrolled action space. Second, evaluation is stage-decomposed into program inference (S1), tool-chain execution (S2), and terminal-state extraction (S3), making it possible to ask *where* a system fails rather than merely whether the final string is correct.

Contributions.

1. **Benchmark for inverse transformation program recovery.** We formulate a multimodal task in which the model must recover a latent transformation program from corrupted video observations and natural-language instructions, then execute the inverse chain and extract the target signal.
2. **Stage-decomposed evaluation.** We decompose the task into program inference (S1), tool-chain execution (S2), and terminal-state extraction (S3), allowing pipeline-stage attribution instead of a single opaque end-to-end score.
3. **Procedural benchmark infrastructure.** CIPHER provides deterministic seed-based generation, contamination resistance, replayable regression cases, and a calibrated difficulty ladder over visibility, chain length, instruction ambiguity, and distractor density. We additionally introduce a video-temporal split-key domain over DAVIS-2017 clips for fast iteration on temporal aggregation and operation inversion.
4. **Evidence of an execution gap.** Across OCR baselines, heuristic pipelines, and frontier multimodal models, strong S3-only extraction does not translate into reliable recovery. The current video-temporal slice further shows that many failed answers contain meaningful partial progress, exposing a gap between perception, temporal evidence selection, and executable policy competence.

2 Task Definition

Setting. Let $\mathcal{V}$ be the space of video clips and $\mathcal{L}$ the space of natural-language strings. CIPHER evaluates a function $f : \mathcal{V} \times \mathcal{L} \rightarrow \Sigma^7$ that maps a corrupted video v' and an instruction $\mathcal{I}$ to a predicted target string $\hat{k}$. The original clip v contains a latent key $k^* \in \Sigma^7$ embedded during a temporal window $[t_s, t_e]$. A sequence of transformation operations $\mathbf{o} = (o_1, \dots, o_n)$ is applied to v to yield $v' = o_n \circ \dots \circ o_1(v)$. The agent’s objective is to recover the inverse program $\mathbf{o}^{-1}$ from $\mathcal{I}$ and v' , execute it, and extract k^* from the recovered terminal state.

The key k^* is drawn from unambiguous alphanumerics ($\Sigma = \text{A-Z} \cup \text{2-9} \setminus \{\text{O}, \text{0}, \text{I}, \text{1}\}$). Instructions are generated from $\mathbf{o}$ via paraphrase templates.

Three-stage pipeline. This task is intentionally composite. An end-to-end failure ($\hat{k} \neq k^*$) may arise because the model inferred the wrong program, realized the wrong tool chain, or failed to localize and read the key after successful recovery. By enforcing a three-stage pipeline—Instruction Parsing (S1), Execution (S2), and Key Extraction (S3)—CIPHER isolates these failure modes. In policy terms, S1 corresponds to *program inference*, S2 to *tool-chain realization*, and S3 to *terminal-state extraction*. This decomposition prevents monolithic metrics from masking which part of the reasoning policy is actually failing.

S1 Parse: $\mathcal{I} \rightarrow \hat{\mathbf{o}}$. Score: sequence-aware op F1.

S2 Execute: run `ffmpeg`($\hat{\mathbf{o}}$). Score: binary validity.

S3 Extract: read $\hat{k}$ from edited video. Score: CharF1 + EM.

Operations. The operation vocabulary is intentionally minimal: six composable `ffmpeg` transforms, `reverse`, `hflip`, `vflip`, `rotate90`, `rotate270`, and `speed2x`. These are reversible, visually

interpretable, and cover both spatial and temporal recovery. The restricted vocabulary is a design choice: it keeps the benchmark focused on inverse program recovery rather than on open ended action-space exploration.

Video-temporal diagnostic domain. In addition to the original inverse-ffmpeg pipeline, we instantiate a controlled video-temporal split-key domain for faster iteration on temporal aggregation. A key is split into two fragments that appear on two consecutive frames of a short clip sampled from the public DAVIS-2017 video corpus [Pont-Tuset et al., 2017]. The solver must identify the target frames, read both fragments, and invert a small symbolic operation before answering. The operation set is `none`, `swap_fragments`, `reverse_chars`, and `swap_reverse_chars`; the last two require character-level transformation rather than simple OCR.

This domain deliberately separates *temporal evidence acquisition* from generic video editing: the latent program, target frame indices, fragment bounding boxes, displayed fragments, normalized fragments, and final answer are all known from generation. This enables dense process scoring. Besides strict EM, we assign partial credit for operation recognition, target-frame localization, displayed-fragment similarity, normalized-answer similarity, transform-derived answer similarity, final-answer similarity, and successful tool use. The resulting traces are used both for model diagnosis and for evolutionary seed selection.

3 The CIPHER Benchmark

CIPHER is designed as a benchmark for inverse procedural reasoning rather than as a static collection of adversarial examples. The generator exposes independent stress axes for policy recovery, while deterministic seeds make every failure replayable.

3.1 Difficulty levels

Table 1: CIPHER difficulty levels (50 samples each, $N=250$ total).

Level	Key vis.	Ops	NL amb.	Distr.	Primary challenge
L1	1 (large)	1	1	0	Single-op sanity check
L2	1 (large)	2	1	0	Two-op program recovery
L3	2 (medium)	2	2	1	Paraphrase + distractor pressure
L4	2 (medium)	3	2	2	Longer chain + distractors
L5	3 (small)	4	3	3	Compositional full difficulty

Key visibility varies font size (13%→7% of frame height), alpha (255→200), and position randomization (centered→random). The key is always rendered in bright yellow with a black drop shadow; distractors use non-yellow colors and are always smaller and dimmer.

Capability axes. The benchmark factors difficulty into independent axes that correspond to distinct stresses on the recovery policy: key visibility controls terminal extraction difficulty; operation count controls latent-program length; instruction ambiguity controls inference difficulty; and distractor density controls false terminal-state attraction. This design supports attribution of performance drops to specific parts of the policy rather than to a single undifferentiated “harder” setting.

NL ambiguity levels. (1) exact canonical phrasing; (2) paraphrase from a pool of templates per operation; (3) compositional “undo” framing with embedded temporal cues. All phrases are intentionally designed to remain scoreable for S1 baselines.

Operation vocabulary. The operational vocabulary consists of six composable transformations: `reverse`, `hflip`, `vflip`, `rotate90`, `rotate270`, and `speed2x`. These operations are visually distinct, non-destructive, and composable via standard `ffmpeg` filters. The vocabulary is intentionally small to preserve interpretability.

3.2 Procedural generation and dataset integrity

Samples are fully deterministic from $(seed, level)$. Background clips are drawn from a Kinetics-400 pool, and the manifest stores the generated sample metadata explicitly.

A persistent challenge in evaluating large language and vision models is the risk of data contamination, where static benchmark data leaks into the pre-training corpus. CIPHER mitigates this through programmatic dataset synthesis. The benchmark is generated from $(seed, level)$, which deterministically selects the background clip, random key, operator chain, and instruction template. Because samples are constructed on demand using disjoint background pools, researchers can regenerate the benchmark with novel seeds, guaranteeing an uncontaminated evaluation environment and a renewable supply of test instances. The same mechanism also makes failures replayable: a single seed is sufficient to reconstruct the exact corrupted observation, latent program, and target timing window for regression-driven debugging.

3.3 Difficulty calibration via GEPA

A benchmark that either saturates (all models at 100% EM) or collapses (all at 0% EM) provides little discriminative signal. Rather than targeting a single operating point, we calibrate a ladder of five extraction conditions, L1–L5, such that a reference frontier VLM spans the full accuracy range from near-ceiling to near-floor. This design supports informative scaling curves and helps preserve discriminative power as models improve.

To find the five conditions we adopt a GEPA-style [Khattab et al., 2025] reflective evolutionary loop over the image-mode generator. This search calibrates *benchmark difficulty*, not model weights. The generator is an S-expression DSL with tunable numerical parameters including key alpha, font size, distractor count and color, Gaussian noise, and blur radius. Each *program* is a fixed instantiation; *fitness* is the squared error between observed EM and a per-level target profile.

We intentionally run this search first in image mode rather than in the full video pipeline. Direct video-level calibration mixes extraction difficulty with pipeline noise from parsing, tool execution, and temporal localization. Image mode yields a cleaner fitness signal for terminal-state readability, after which the resulting ladder can be transferred into the full procedural benchmark.

Table 2: Calibrated extraction levels, measured on Claude Sonnet 4.6 (reference model during calibration, $n=8-16$ per level).

Level	Config (α , font%, dist, σ , blur)	target EM	obs. EM	char-acc
L1	255, 12%, 0, 0, 0	95%	62%	94%
L2	240, 10.5%, 0, 0, 0	70%	69%	94%
L3	225, 9%, 0, 0, 0	40%	44%	92%
L4	215, 8%, 1c, 0, 0	15%	25%	82%
L5	205, 7.5%, 1 (1y), 0.5, 0	3%	0%	55%

3.4 Metrics

EM. $\frac{1}{N} \sum_i \mathbf{1}[\hat{k}_i = k_i^*]$.

CharF1. Character-level precision/recall F1 for near-miss credit.

PipeF1.

$$\text{PipeF1} = 0.25 \cdot \overline{S1} + 0.25 \cdot \overline{S2} + 0.50 \cdot \overline{\text{CharF1}} \quad (1)$$

S3 receives double weight as the terminal objective; S1 and S2 are diagnostic leading indicators for policy failure attribution.

4 Experiments

4.1 Experimental setup

Inference modes. We evaluate models in two settings: (1) **S3-only**: models receive a ground-truth corrected frame containing the key and must read it. This tests pure terminal-state extraction and serves as an upper-bound control slice. (2) **Full pipeline**: models receive the scrambled .mp4 video and natural-language instructions with Bash access, and must independently parse instructions (S1), execute `ffmpeg` corrections (S2), locate the key window, and extract the key (S3).

The distinction is central: strong performance in S3-only does not imply strong inverse program recovery. It only shows that once the correct terminal state is provided, the model can decode the target string. Full-pipeline performance measures whether the system can actually reach that terminal state.

Agentic evaluation. For full-pipeline evaluation, VLM agents are given Bash access and must decide their own video processing strategy (`ffmpeg` flags, frame sampling, temporal search). No ground-truth metadata (operations, key, timing) is provided.

Reported metrics. We report strict exact match (EM), character-level F1, stage-specific S1 parse F1, S2 execution success rate, and Pipeline-F1. In the current draft, the strongest supported evidence comes from S3-only and calibrated extraction experiments; full-pipeline reporting remains incomplete and is therefore presented as an explicit gap rather than as a filled leaderboard.

For the video-temporal diagnostic domain, we additionally report *process partial credit*. This score decomposes a failed answer into operation credit, frame-localization credit, displayed-fragment similarity, normalized-answer similarity, transform-derived-answer similarity, final-answer similarity, and tool-use credit. It is intentionally not a replacement for EM; it is a dense diagnostic signal used to distinguish uninformative failures from meaningful near misses.

4.2 Models

We benchmark OCR baselines, heuristic pipelines, and frontier multimodal systems:

OCR-only Pure extraction baseline. Uses `easyocr` on sampled key frames; no language understanding, program inference, or tool use.

Yellow-OCR Color-filtered OCR baseline. Isolates yellow pixels via HSV masking before running `easyocr`. Tests whether color-aware preprocessing can substitute for reasoning or executable recovery.

MCP Skill Heuristic agentic baseline (`solve_captcha.py`): regex-based instruction parsing (S1), `ffmpeg` execution (S2), and a multi-method extraction chain (S3).

Claude Haiku / Sonnet / Opus API-first frontier multimodal systems used to probe the gap between prepared extraction and end-to-end recovery.

GPT-5.4 Tool-using frontier model evaluated through the same full-pipeline protocol.

4.3 Current results

Table 3: S3 key extraction from ground-truth corrected frames ($n=5$ per level). Best per column in **bold**.

Model	L1	L2	L3	L4	Total
OCR-only	20%	25%	30%	25%	25%
Yellow-OCR	33%	—	—	—	—
MCP Skill	20%	25%	30%	25%	25%
Haiku	100%	60%	40%	100%	75%
Sonnet	100%	100%	100%	100%	100%
Opus	100%	100%	100%	100%	100%

Table 4: Full pipeline results (video $\rightarrow$ key). This table is intentionally left incomplete in the current draft; filling it is a required next experiment because the benchmark’s main claim concerns the gap between S3-only extraction and end-to-end recovery.

Model	L1 EM	L2 EM	L3 EM	L4 EM	EM (all)	PipeF1
MCP Skill	–	–	–	–	–	–
Haiku	–	–	–	–	–	–
Sonnet	–	–	–	–	–	–
Opus	–	–	–	–	–	–
GPT-5.4	–	–	–	–	–	–

Overview. Table 3 reports S3 extraction accuracy in isolation: each model receives a corrected frame containing the yellow key and must read it. Frontier VLMs (Opus, Sonnet) achieve perfect accuracy, while Haiku drops on smaller, randomly positioned keys. OCR baselines perform poorly due to background text interference and clutter.

Crucially, S3 in isolation is *not* the binding bottleneck for frontier models. The benchmark’s real difficulty lies in the full recovery policy: agents must infer the inverse transformation program, execute it correctly, and identify the correct temporal window containing the key. The missing full pipeline table is therefore not a cosmetic omission; it is the central remaining experiment required to validate the benchmark’s main claim.

4.4 Video-temporal diagnostic results

We also ran a controlled DAVIS-2017 video-temporal split in which the model must aggregate two consecutive frames and invert a symbolic fragment transformation. This split is not yet the full inverse-fmpg benchmark; it is the current fast iteration domain for studying temporal aggregation, solvability, and dense process metrics. The current pool contains 78 generated tasks with process metrics and 233 model responses across three CLI backends. All modern samples in this slice use dynamic DAVIS frame sequences and pass the current “likely-readable” solvability gate.

Table 5: Current DAVIS video-temporal diagnostic slice. EM is strict exact match. Missing counts empty/NONE final answers. Partial is dense process partial credit. Values are means over response rows; $\pm$ is population standard deviation.

Backend	n	EM $\uparrow$	Missing $\downarrow$	Partial $\uparrow$
Claude CLI Haiku	78	14.1 $\pm$ 34.8	1.3 $\pm$ 11.2	43.1 $\pm$ 28.6
Codex CLI GPT-5.3 Spark	77	0.0 $\pm$ 0.0	100.0 $\pm$ 0.0	6.5 $\pm$ 5.6
Gemini CLI Flash Lite	78	0.0 $\pm$ 0.0	98.7 $\pm$ 11.2	5.0 $\pm$ 5.7

At the task level, 11 of 78 tasks are *strict-good*: at least one model achieves EM. A larger set of 45 tasks is *semantic-good*, meaning either EM or maximum process partial credit at least 0.25. Finally, 77 tasks have at least one non-missing response with partial credit at least 0.15, making them usable for debugging even when no backend solves them exactly. The gap between strict-good and semantic-good is the most important observation: the split contains many failures that are not random, but partially correct trajectories.

The strongest current backend is Claude CLI Haiku, which often identifies the operation and produces structured intermediate evidence but still loses characters or over-reads distractors on hard samples. Codex Spark and Gemini Flash Lite mostly fail by returning no final answer in this setup, although their traces occasionally receive small operation/tool credit. This suggests that backend/tool framing and response protocol are part of the benchmark surface, not merely implementation details.

4.5 Required experiments for the camera-ready version

The current draft still lacks the analyses that best express CIPHER’s identity as a diagnostic benchmark:

- scaling curves over chain length, instruction ambiguity, distractor density, and temporal sparsity;
- paired S3-only vs full-pipeline comparisons with an explicit execution-gap metric;
- stage-attribution plots for S1, S2, and S3 across model families; and
- a curated failure-seed suite with traces.

The video-temporal slice supplies the first version of this failure-seed suite: 12 seedbank genomes were selected by validity, non-missing rate, process partial credit, frame-localization credit, transform similarity, final-answer similarity, and diversity. The selected genomes cover all four fragment operations and range from easy validity-tuned settings (4–6 frames, low noise, large bright text) to hard-mined settings (7–8 frames, high jitter, distractors, and `swap_reverse_chars`).

5 Analysis

CIPHER is most valuable when failures are treated as structured signals rather than as a flat collection of incorrect strings. The analysis should therefore be read as a *failure geometry*: which stage breaks first, how errors change with difficulty, and which seeds should be preserved as regression cases. Some subsections below remain partially prospective pending full-pipeline results, but the intended analytical frame is already clear.

5.1 Stage bottleneck analysis

We decompose failures by stage to identify where the recovery policy breaks down.

S1 (program inference). On simple levels, S1 often saturates first: regex matching achieves high parse F1 on canonical instructions. The remaining errors are not random; they concentrate in paraphrastic or compositional instructions where the system must recover the latent operator sequence rather than match surface forms.

S2 (tool-chain execution). Execution failures are rare on short chains but increase with operation count, primarily because the model must realize a valid `ffmpeg` chain rather than merely name the right operations.

S3 (terminal-state extraction). S3 is *not* the binding bottleneck for frontier VLMs once the correct frame is provided. The true difficulty lies in the coupling between temporal localization and execution: many systems that can read the key still fail to arrive at the correct frame.

Near-miss errors are informative. Near-miss predictions should not be discarded as noise. They reveal whether a model reached the correct terminal state but misread one character, localized the wrong frame, or recovered the wrong but visually plausible distractor. In a diagnostic benchmark, these distinctions matter more than a single binary EM label.

5.2 Character-level confusion in the near-miss regime

Under calibrated extraction settings, models produce consistent substitutions among visually similar glyphs rather than random strings. These errors are useful because they identify a readable-but-fragile regime in which the task is still discriminative.

5.3 Video-temporal failure taxonomy

The DAVIS video-temporal slice makes the failure geometry concrete. Among the 78 modern video-temporal tasks, 67 currently have no exact match from any evaluated backend, but only one task is an all-missing failure across all models. This is the desired hard-but-diagnostic regime: most failures still contain usable intermediate evidence.

The dominant failure families are:

- **near misses:** 21 tasks have no EM but reach maximum process partial credit at least 0.5, indicating that a model recovered part of the operation/frame/fragment structure but failed the final string;
- **frame-localization weakness:** 24 tasks have maximum frame-localization credit below 0.5, showing that models often read from non-target frames or from the entire clip rather than the two target frames;
- **over-reading:** 12 tasks contain predictions longer than the gold key by more than two characters, usually because the model absorbs distractors or neighboring yellow text into the answer; and
- **hard swap-and-reverse cases:** the highest-difficulty frontier is dominated by `swap_reverse_chars` seeds with 7–8 frames, high jitter, distractors, and non-central placement.

This taxonomy directly motivates the dense metric. A flat EM score would mark all of these examples as identical failures. Process partial credit separates “no evidence”, “found the operation”, “localized the wrong frame”, and “read the right evidence but transformed it incorrectly”, which are different failure modes for an agentic multimodal system.

5.4 Canonical failure suite

For benchmark-driven development, we recommend preserving a curated suite of 20–50 canonical failure seeds with associated traces. Each entry should store: seed, level, background regime, gold operator chain, predicted operator chain, execution trace, temporal localization decision, predicted key, and error family. This turns CIPHER from a static benchmark into a replayable regression environment for agentic systems.

The current video-temporal seedbank is the first implementation of this idea. It stores not only the winning candidates, but also diverse non-winning candidates selected by a utilization–diversity criterion. This matters because the most useful benchmark samples are not always the hardest candidates by EM: some are slightly easier but much more interpretable, and some expose specific operation or temporal-localization weaknesses that would be lost if we kept only the scalar best-of-run candidate.

6 Conclusion

We introduced CIPHER, a procedurally generated benchmark for inverse transformation program recovery in tool-using multimodal systems. CIPHER asks whether a model can infer, invert, and execute a latent transformation program from corrupted video observations and natural-language instructions, then localize and extract the target signal. Its three-stage decomposition and Pipeline-F1 metric enable fine-grained diagnosis of where the recovery policy breaks down.

Our current results already show a strong asymmetry: frontier VLMs solve S3 key extraction near-perfectly in isolation, while OCR baselines struggle. The more important open question, and the central claim CIPHER is designed to measure, is the gap between this prepared extraction competence and reliable end-to-end policy execution under temporal search and tool use. The benchmark is therefore most useful not as a flat leaderboard, but as a diagnostic environment for stage-attributed failure analysis and reproducible regression testing.

The new DAVIS video-temporal slice strengthens this claim. Even before the full inverse-ffmpeg table is complete, the split-key domain shows that strict EM alone hides useful structure: many samples are not solved exactly, yet model traces contain partially correct operation, frame-localization, or fragment-normalization evidence. This supports CIPHER’s core design principle: the benchmark should preserve process traces and dense diagnostics rather than compress all behavior into a single final-string score.

Limitations. CIPHER focuses on a small deterministic ffmpeg vocabulary; generative edits and audio-only keys are out of scope. Human baselines are not included in this version.

Future work. The most important next steps are empirical rather than nominal: complete the paired S3-only vs full-pipeline table, add scaling curves over chain length, instruction ambiguity, distractor density, and temporal sparsity, and curate a canonical failure-seed suite with traces. Beyond that, temporal edit expansion, multi-frame extraction, and stronger policy-search baselines are natural extensions.

For the video-temporal domain specifically, the immediate target is a frozen pool of roughly one hundred audited valid samples, balanced across none, `swap_fragments`, `reverse_chars`, and `swap_reverse_chars`. This pool should be evaluated with Haiku/Sonnet/Opus, Codex, and Gemini backends under paired RPAD-style debiasing so that hardness estimates are not dominated by one model family’s response format or tool-use behavior.

Reproducibility. All code and generation scripts are released at <https://github.com/Fr0do/cipher>. The benchmark generator is deterministic by seed and level, enabling replay of individual failure cases.

Acknowledgements

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